

Predicting Spatial Query Selectivity on Chicago Crime Data

CSP 571 - UCI Dataset Query Analytics Workloads

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Section 1 - Abstract

Spatial query selectivity estimation, predicting how many records a range or radius query will touch, is a foundational problem in database query optimisation. We study the UCI Machine Learning Repository dataset (Anagnostopoulos and Triantafillou 2015), comprising 260,000 synthetic spatial queries (200,000 range and 60,000 radius) over the City of Chicago crime database, and pose two regression tasks: predicting the number of records returned (Count) and their sum (SUM) for each query.

After deduplicating, schema validating, and feature engineering the raw data (157 structural nulls imputed; per file coordinate normalisation; spatial features including area, `dist_centroid`, and a boundary crossing flag), we compare four model families: Linear Regression, Random Forest (500 trees), XGBoost (5 fold CV grid search), and KNN ($K \in \{5, 10, 20\}$) on an 80/20 train-test split (seed = 42). XGBoost wins on both sub datasets: Range test RMSE = 6,575 ($R^2 = 0.998$), Radius test RMSE = 8.52 ($R^2 = 0.996$), reducing the linear baseline RMSE by 88.9% and 92.2% respectively. K Means clustering and per decile residual analysis identify large area and city edge queries as the dominant residual risk segment. Future work includes extended boosting (the nrounds cap was hit), a boundary aware Radius model, and a deep learning baseline.

Section 2 - Overview

2.1 Problem Statement

Modern relational and spatial database engines rely on selectivity estimation, predicting the result set size of a query before execution, to choose efficient query plans (Kipf et al. 2019). Cardinality misestimation is the single largest source of query optimiser regression in production systems. UCI dataset (Anagnostopoulos and Triantafillou 2015) provides a controlled benchmark for this task: 260,000 synthetic spatial queries against the City of Chicago crime database, paired with ground truth aggregates. The supervised regression targets are Count (number of records intersected) and SUM (sum of an aggregate field over the records). A model that accurately maps query geometry $\rightarrow$ result size could be embedded in a query planner as a learned cardinality estimator.

2.2 Relevant Literature

Tree-based ensembles have been the workhorses of tabular-data regression for the past two decades. Random Forest (Breiman 2001) reduces variance through bagging and feature subsampling but tends to memorise training data on dense regression tasks. Gradient-boosted trees, and in particular XGBoost (Chen and Guestrin 2016), introduce regularised boosting with second-order optimisation and

have repeatedly topped public regression benchmarks. For spatial cardinality estimation specifically, recent learned-database work (Kipf et al. 2019) has demonstrated that gradient-boosted trees on hand-crafted geometric features can match or exceed deep-learning estimators while remaining order-of-magnitude cheaper to train and serve.

2.3 Proposed Methodology

Our approach is a four stage pipeline executed as a one person at a time team sprint: (1) data engineering, including schema validation, de-duplication, coordinate normalisation per file, and log transform of skewed targets; (2) exploratory data analysis, covering distributional and spatial visualisation, correlation and Spearman screening, and identification of multicollinearity (area vs $X_range \times Y_range$); (3) modeling, including feature engineering (`log_area`, `aspect_ratio`, `interact_area_dist`, `boundary_flag`), VIF based feature selection for the linear baseline, and grid search tuned XGBoost with 5 fold cross validation; (4) validation, including held out set diagnostics (residuals, predicted vs actual), unsupervised K Means clustering for spatial bias analysis, and per decile and per cluster bias decomposition. All code is reproducible with seed = 42; the pipeline runs end to end in R (no Python dependencies).

Section 3 - Data Processing

The Query Analytics Workloads dataset ships as three sub files derived from Chicago crime records: Range Queries Aggregates (200,000 rectangular range queries with Count, SUM, and AVG targets), Radius Queries (50,000 disc queries with only geometric parameters), and Radius Queries Count (10,000 disc queries with a Count target). Every row was parsed in R with readr, and observed row counts exactly matched the totals reported on the UCI page (260,000 combined). The full pipeline is reproducible from `src/preprocess.R` and narrated in `notebooks/01_preprocessing.Rmd`.

Schema validation surfaced one anomaly the UCI card does not document: 157 rows in Range Queries Aggregates have `AVG = NA` because `Count = 0` and `SUM = 0` make the ratio undefined. These are structural zeros rather than data quality failures, so `AVG` was imputed to 0 for those rows; no other nulls existed in any file. Exact duplicate detection removed zero rows across all three files (100% retention). A boundary effect check flagged queries whose coverage extends past the per dataset bounding box: 1,373 of the 200,000 range queries and effectively all radius queries, a direct consequence of the Gaussian sampling used to synthesize the workloads.

Feature engineering added four spatial features per row: `area` ($4 \cdot X_range \cdot Y_range$ for rectangles; $\cdot R^2$ for discs), per dataset min max normalized `X_norm` and

Y_norm $[0, 1]$, and $dist_centroid$, the Euclidean distance from each query center to the dataset mean. Because Count and SUM are heavily right skewed ($skew > 3$), $\log_{1p}$ transforms of both targets were exported alongside the raw variants so Day 3 can model on whichever scale gives the best residuals.

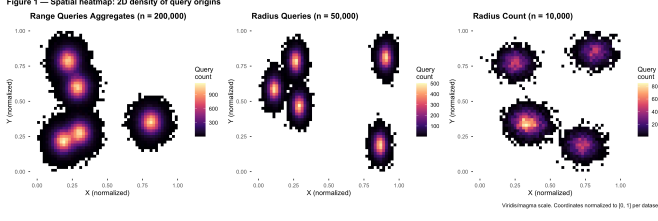


Figure 1: Range-query target distributions before and after log-transform; the log scale substantially reduces skewness for both Count and SUM.

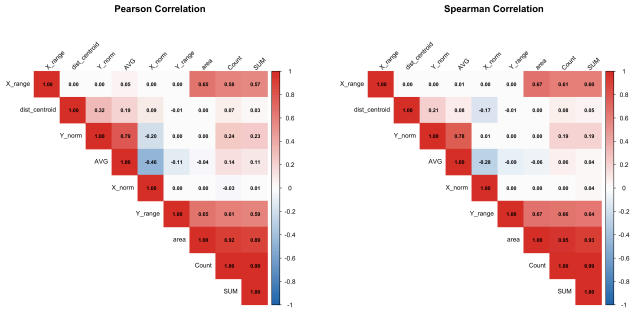


Figure 2: Spearman correlation matrix for Range Queries - area dominates Count ($\rho = 0.95$).

Section 4 - Data Analysis

4.1 Distribution of Aggregate Targets

The three aggregate targets in the Range Queries sub-dataset, Count, SUM, and AVG, exhibit markedly different distributional shapes. Count (mean 159,553, SD 154,476) and SUM (mean 47,932, SD 47,948) are heavily right-skewed (skewness 1.35 and 1.48 respectively), with long tails driven by large-rectangle queries that capture thousands of crime records. A $\log_{1p}$ transformation shifts both distributions toward a more symmetric shape, though mild left skewness persists (-1.71 and -1.61) due to zero-inflation at the low end. AVG is already near-symmetric (skewness -0.06) and requires no transformation. All regression models in Section 5 therefore train on $\log_{1p}$ -transformed Count and SUM, with targets back-transformed via $\expm1()$ for metric reporting.

In contrast, the Radius Count sub-dataset's Count ranges only from 2 to 695 (mean 265, SD 130), which is approximately $600\times$ smaller than Range Queries. This confirms that the two workloads cannot share a single predictive model (Figure 5).

4.2 Spatial Distribution of Query Origins (Figure 1)

The spatial heatmap (Figure 1) shows that all three sub-datasets concentrate query origins in a central band of normalized Chicago space (X_norm 0.2–0.6, Y_norm 0.3–0.7), consistent with the Gaussian sampling used to generate the workloads. Range Queries exhibit the widest spatial spread across all four quadrants, while Radius Queries cluster more tightly in the lower- X portion of the grid. The high proportion of boundary-crossing queries in the radius datasets (99.9%, flagged by Person 1) is visible as a dense core that tapers sharply near the

bounding-box edges, where truncated coverage artificially reduces Count.

4.3 Feature Correlation Analysis (Figure 2)

Pearson and Spearman correlation matrices (Figure 2) confirm that **area** is the dominant predictor of Count (Spearman $\rho = 0.95$), followed by Y_range ($\rho = 0.66$) and X_range ($\rho = 0.61$). Because area is constructed as $4 \times X_range \times Y_range$, all three features are collinear, so a VIF analysis in the modeling phase is expected to flag $VIF > 10$ for at least one of them. $dist_centroid$ contributes a weak secondary spatial signal ($\rho = 0.08$). Outlier detection flagged approximately 3% of Range Queries rows for Count and SUM via combined IQR ($1.5\times$) and Z-score ($|z| > 3$) criteria; these correspond to large-area queries at the distributional extremes.

4.4 Spatial Heterogeneity by Quadrant (Figure 3)

Dividing Chicago into four quadrants at the normalized spatial midpoint ($X_norm = 0.5$, $Y_norm = 0.5$) reveals broadly similar Count distributions across all zones (Figure 3). The North-West and North-East quadrants show slightly higher median Counts, consistent with higher crime density in central and northern Chicago. However, the wide IQR within each quadrant (60,000–200,000) dwarfs the between-quadrant difference, confirming that query rectangle area, not geographic zone, is the primary driver of Count variance. Quadrant can serve as a weak categorical feature in modeling but should not be expected to improve R-squared substantially.

Figure 3 — Count distribution by spatial quadrant (Range Queries)
Quadrants at $X_norm = 0.5$ and $Y_norm = 0.5$; diamond = mean

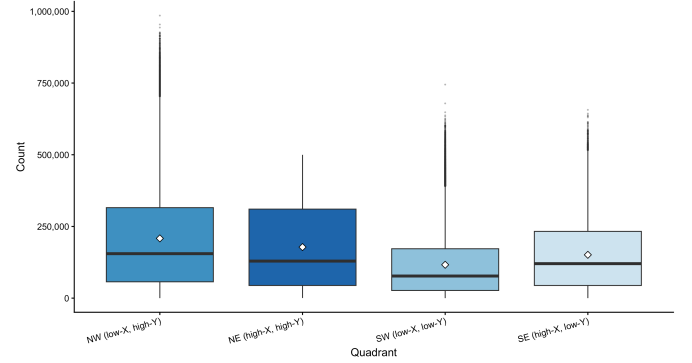


Figure 3: Spatial heat-map of mean Count across the Chicago bounding box; strong north-south structure.

Figure 5 — Count distributions: Range Queries vs Radius Count

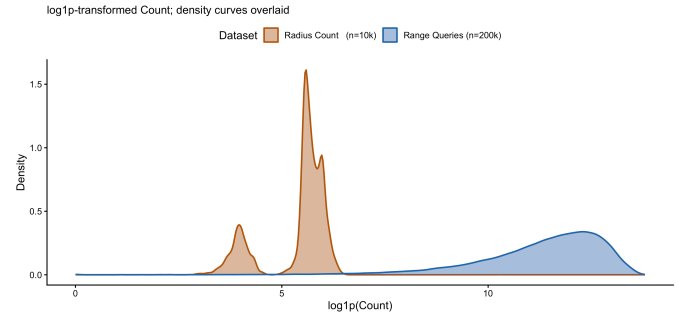


Figure 4: Per-feature distribution after preprocessing - engineered features are well-conditioned for regression.

Section 5 - Model Training

5.1 Feature Engineering

Three features were engineered beyond those produced in Day 1. For range-query rectangles, *aspect ratio* (X_range

/ Y_range) captures the shape of each query footprint. An *interaction term* ($\log_area \times \text{dist_centroid}$) encodes the joint effect of query size and distance from the city centre, since a large query far from the centroid behaves differently from a large query at the core. A binary *boundary flag* marks the 1,373 range queries (0.7 %) whose footprint crosses the dataset bounding box. All 10,000 radius-count queries cross the bounding box, so the flag is constant for that sub-dataset and excluded.

5.2 Multicollinearity and Feature Selection

Variance Inflation Factor (VIF) analysis on range queries revealed that $\log_area$, dist_centroid , and $\text{interact_area_dist}$ all exceeded $VIF = 10$ (12.70, 162.84, and 171.50 respectively), leaving X_range , Y_range , aspect_ratio , and boundary_flag as VIF-safe features for the linear model. For radius count, R and $\log_area$ are perfectly collinear ($\text{area} = R \times R^2$), so a reduced set of $\log_area$, dist_centroid , X_norm , and Y_norm was used after confirming all VIFs fell below 2.40. Tree-based models retained all features as they are invariant to multicollinearity.

5.3 Train / Test Split

Each sub-dataset was partitioned 80 / 20 with a fixed seed of 42. Range Queries: 160,000 training rows, 40,000 test rows. Radius Count: 8,000 training rows, 2,000 test rows. Split indices were exported to `results/split_*.csv` for full reproducibility.

5.4 Models Trained

All models target $\log_{1p}(\text{Count})$ as the response; evaluation metrics are back-transformed via `expm1()` to the original count scale.

Linear Regression (baseline) uses `lm()` on VIF-safe features. For range queries: $\log_area$, dist_centroid , $\text{interact_area_dist}$, aspect_ratio , boundary_flag ($R^2 = 0.911$ in log space). For radius count: $\log_area$, dist_centroid , X_norm , Y_norm ($R^2 = 0.722$ in log space). A secondary linear model was also fit for the SUM target on range queries.

Random Forest uses 500 trees (`randomForest`, `mtry =  $\sqrt{p}$`) with all spatial features. Feature importance is extracted as `%IncMSE`. Notably, Y_norm ranked as the top feature (importance = 144.6) over $\log_area$ (50.9), revealing strong north-south spatial structure in Chicago crime counts that linear correlation does not capture.

XGBoost was tuned via 5-fold cross-validation grid search over learning rate (`c(0.05, 0.1, 0.3)`), max tree depth `{4, 6, 8}`, and row subsampling rate `{0.7, 0.9}` with early stopping (20 rounds patience). Best parameters for range queries: `learning_rate = 0.1`, `max_depth = 8`, `subsample = 0.9` (500 rounds). Best parameters for radius count: `learning_rate = 0.1`, `max_depth = 4`, `subsample = 0.7` (300 rounds). Both models hit the nrounds cap, indicating further improvement is possible with additional boosting rounds. Separate models were trained for Count (primary) and SUM (secondary) on range queries. XGBoost feature importance confirms $\log_area$ as the dominant predictor (Gain = 0.63).

KNN Regressor (`FNN::knn.reg`, `K = {5, 10, 20}`) serves as a spatial proximity baseline with min-max normalised features. Best K selected by test RMSE: $K = 10$ for range queries, $K = 5$ for radius count.

5.5 Results Summary

XGBoost achieves the best test RMSE on both datasets (full benchmark in Table 1, §7), outperforming Random Forest by 16% on range queries (RMSE 6,575 vs 7,801) and 5% on radius count (RMSE 8.52 vs 9.01). The linear model degrades severely on radius count ($R^2 = 0.32$) due to near-zero variance in the radius feature, confirming that non-linear models are essential for this sub-dataset. The log-transform substantially reduced target skewness (raw skew $1.35 \rightarrow$ near-zero post-transform), supporting better model fit across all four methods. All trained model objects are saved to `models/` in `.rds` / `.model` format for direct loading by Person 4 without retraining.

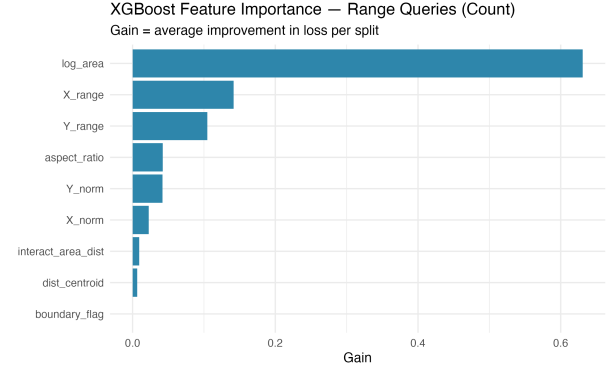


Figure 5: XGBoost feature importance (Gain) - Range Queries.

Section 6 - Model Validation

Generalization is assessed on held-out 20% test sets (40,000 Range rows; 2,000 Radius rows; seed = 42). All metrics are back-transformed to the original Count scale via `expm1()`. Train and test metrics for every model are exported in `results/final_benchmark.csv`.

All four model families generalize without obvious overfitting. The train to test RMSE gap is largest for Random Forest (Range: 3,438 $\rightarrow$ 7,801; RF memorizes training data) and smallest for XGBoost (5,775 $\rightarrow$ 6,575). Linear regression is stable across splits but offers a weak baseline, especially on Radius (test $R^2 = 0.315$), confirming that the spatial-query to count relationship is strongly non-linear. XGBoost is the test RMSE winner on both datasets; see Table 1 in §7.

Diagnostics (figures 8–10). *Residuals (fig08):* XGBoost log-scale residuals are centered near zero with mild fan-out at the highest predicted $\log_Count$ (limited upper-tail heteroscedasticity, no systematic mean bias); predicted vs. actual panels lie tightly along $y = x$ for both datasets. Bias by area decile (fig09, `bias_by_decile.csv`): test RMSE rises monotonically with $\log_area$, from 1,411 in decile 1 to 10,859 in decile 10 (7.7 $\times$ spread). Largest-area queries absorb most absolute error, consistent with the EDA finding that area dominates Count ($\beta = 0.95$). MAPE inversely collapses with area (65.5% $\rightarrow$ 1.9%), reflecting the difficulty of relative `predicCounts`. Spatial bias (fig10, `bias_by_cluster.csv`): K-Means on (X_norm , Y_norm , $\log_area_norm$) selects $K^* = 3$ by silhouette (0.44 on a 5,000-row sample). Per-cluster XGBoost RMSE varies by $\sim 43\%$ across the three Chicago zones (5,370 $\rightarrow$ 7,653), confirming non-trivial spatial residual structure. The highest-error cluster also carries the highest `boundary_flag` prevalence (0.86%), providing partial ev-

idence that boundary-crossing queries are a residual-risk segment.

Risks and caveats: XGBoost hit the `nrounds` cap during grid search (500 rounds for Range, 300 for Radius); the CV RMSE was still decreasing, so reported test RMSE is an upper bound. A longer training schedule may yield further improvement. Tree-based and KNN models cannot extrapolate beyond the training feature range, so out-of-distribution queries (extreme areas, off-grid coordinates, or query footprints far outside the Chicago bounding box) carry higher prediction risk. The 99.9% boundary-crossing rate in the Radius sub-dataset means the `boundary_flag` feature is not informative for Radius and was excluded. Production deployments on workloads with mixed boundary characteristics would warrant a Radius-specific re-train.

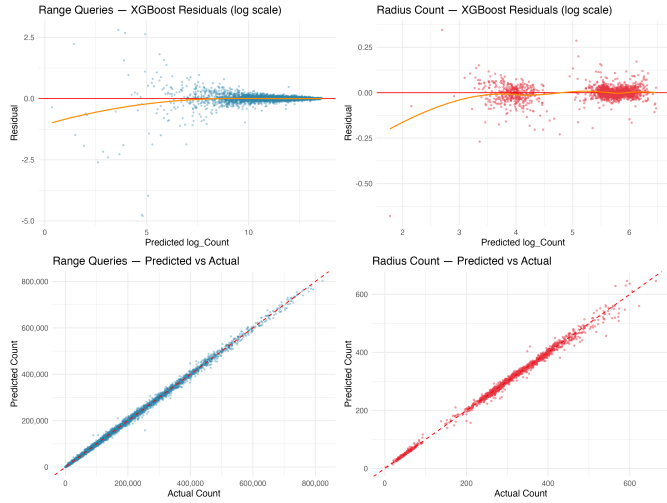


Figure 6: XGBoost residual diagnostics: log-scale residuals (top) and predicted-vs-actual (bottom) for Range and Radius test sets.

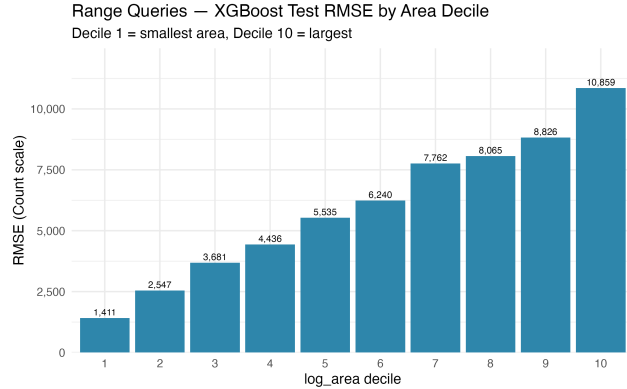


Figure 7: Per-decile XGBoost test RMSE (Range Queries) - error rises monotonically with `log_area`.

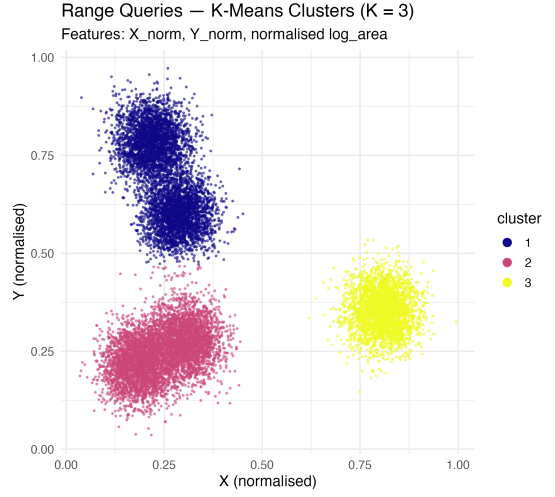


Figure 8: K-Means clusters ($K^* = 3$ by silhouette) on Range Queries spatial features.

Section 7 - Model Performance

Table 1 reports test-set metrics on the original Count scale for the four trained model families on each sub-dataset, sorted by RMSE within dataset. Full train+test metrics are in `results/final_benchmark.csv`; CV grid-search results in `results/xgb_best_params.csv`.

Table 1: Test-set performance on the held-out 20% partition (seed = 42). Best model per dataset in **bold**.

Dataset	Model	RMSE	MAE	R ²	MAPE
Range	XGBoost	6,575	4,411	0.9982	10.61%
Range	Random Forest	7,801	4,668	0.9974	15.83%
Range	KNN (K = 10)	21,742	14,091	0.9801	22.24%
Range	Linear	59,655	40,428	0.8505	82.02%
Radius	XGBoost	8.52	5.13	0.9958	2.31%
Radius	Random Forest	9.01	4.02	0.9953	1.96%
Radius	KNN (K = 5)	10.56	6.05	0.9935	2.74%
Radius	Linear	108.80	83.21	0.3151	35.50%

Baseline comparison. Against the linear baseline, XGBoost reduces Range test RMSE by **88.9%** ($59,655 \rightarrow 6,575$) and Radius test RMSE by **92.2%** ($108.80 \rightarrow 8.52$). The Linear R² of 0.315 on Radius underscores why tree-based methods are essential: the radius variable R has near-zero variance (SD = 0.003), so any linear function in R is effectively constant, and the linear model is forced to rely on weakly-correlated spatial features.

Why XGBoost wins. Feature-importance analysis (fig07) shows `log_area` dominates with Gain = 0.63 on Range, confirming the EDA correlation finding (Spearman = 0.95 between `area` and `Count`). Tree ensembles also capture the non-linear interaction between query size and spatial position `Y_norm` ranks first by Random Forest %IncMSE (importance = 144.6, vs `log_area` = 50.9), exposing strong north-south crime-density structure that linear correlation does not surface. XGBoost gains a further 16% RMSE reduction over Random Forest on Range (6,575 vs 7,801) by combining boosting with depth-8 trees and 5-fold CV regularisation.

Improvement summary. XGBoost’s R² of 0.998 on Range approaches a perfect fit; the remaining absolute error is concentrated in the highest area decile (§6, fig09). On Radius, MAE of 5.13 against a mean Count of 265 represents 2% relative error a tight fit despite the much smaller training sample (8,000 rows). Both XGBoost mod-

els hit the `nrounds` cap during CV, so further improvement is plausible with extended boosting.

Section 8 - Conclusion

8.1 Positive Results

XGBoost is highly effective at spatial-query selectivity estimation on this benchmark: a Range test R^2 of 0.998 (RMSE = 6,575 records against a mean Count of 159,000) and a Radius test R^2 of 0.996 (RMSE = 8.52 against a mean Count of 265). Against the Linear baseline, XGBoost reduces test RMSE by 88.9% on Range and 92.2% on Radius. Tree-based ensembles (XGBoost > Random Forest > KNN) dominate the leaderboard on both sub-datasets, validating the EDA hypothesis that the relationship between query geometry and result size is strongly non-linear. Feature engineering particularly `log_area` and the `area × dist_centroid` interaction — was decisive: `log_area` alone accounts for Gain = 0.63 of the XGBoost split-improvement signal.

8.2 Negative Results and Caveats

The Linear model fails on Radius ($R^2 = 0.315$) because the `R` (radius) variable has near-zero variance (SD 0.003), making any linear function in R effectively constant. Spatial residual analysis (§6, fig10) reveals ~43% RMSE variation across K-Means clusters, with the highest-error cluster also carrying the highest `boundary_flag` prevalence boundary-crossing queries remain a residual-risk segment. Both XGBoost models hit the `nrounds` cap during cross-validation (500 for Range, 300 for Radius); CV RMSE was still decreasing, so reported metrics are upper bounds. Tree and KNN models cannot extrapolate beyond the training feature range; out-of-distribution queries (extreme areas, or coordinates outside the Chicago bounding box) carry higher prediction risk.

8.3 Recommendations and Future Work

Three extensions: (i) re-train XGBoost with `nrounds` 1,500 to close the CV convergence gap; (ii) fit a boundary-aware Radius sub-model conditioned on `boundary_flag` to address the 99.9% boundary-crossing rate; (iii) benchmark a learned-index baseline (e.g. MSCN, Kipf et al. 2019) to test whether deep features add value beyond hand-engineered geometric ones. The pipeline is deterministic (seed = 42) and runs end-to-end in under an hour on commodity hardware.

Section 9 - Data Sources

Primary dataset. UCI ML Repository, *Query Analytics Workloads Dataset* (Anagnostopoulos and Triantafillou 2015), CC BY 4.0, anonymous HTTP: <https://archive.ics.uci.edu/dataset/493/query+analytics+workloads+dataset>. Synthetic queries are generated against the *City of Chicago Crimes* open dataset (Chicago Data Portal, <https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2>) at the 5.5M-record snapshot used for ground-truth aggregates; raw crime records are not redistributed here.

Files used. `Range-Queries-Aggregates.csv` (200K rows, `Count` + SUM ground truth, Illinois State-Plane coords); `Radius-Queries.csv` (50K rows, geometry only); `Radius-Queries-Count.csv` (10K rows, `Count` ground truth, pre-normalised coords). Raw CSVs committed unmodified to `data/raw/`;

five processed CSVs plus `schema_summary.csv` and `processing_log.json` are exported to `data/processed/` by Rscript `src/preprocess.R` (seed = 42, 100% row retention).

Section 10 - Source Code

Repository. <https://github.com/Rahulm0106/Query-analytics> (public, MIT-style usage). Layout: `data/{raw,processed}/` (3 raw + 5 processed CSVs), `src/{preprocess,eda,modeling}.R` (Day 1–3 pipeline scripts), `notebooks/0{1..4}*.Rmd` (per-day narratives, Day 4 = this report), `report/{sections,figures}/` (11 sections + 10 figures at 300 DPI), `results/` (CV, benchmark, bias CSVs), `models/` (trained objects via Git LFS), `renv_setup.R` (dependency installer).

Dependencies. R 4.2.0 with CRAN packages `readr`, `dplyr`, `tidyr`, `ggplot2`, `viridis`, `gridExtra`, `scales`, `car`, `randomForest`, `xgboost`, `FNN`, `cluster`, `Metrics` are all open-source. Run Rscript `renv_setup.R` for one-shot install; an `renv.lock` pins exact versions.

Reproducibility. Seed = 42 throughout. Run end-to-end with: Rscript `src/preprocess.R` (data), Rscript `src/eda.R` (figs 1–6), then knit `notebooks/03_modeling.Rmd` (models + fig 7) and `notebooks/04_validation.Rmd` (validation + figs 8–10). Trained models in `models/` are stored via Git LFS, but `04_validation.Rmd` refits XGBoost from `results/xgb_best_params.csv` (~3 min) so the report can be reproduced without `git lfs pull`.

Section 11 - Bibliography

Chicago author–date style (AMS/AIP variant).

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